GCSE Maths misconception bank

Fractions, algebra, graphs, shape, probability, and the formulas that get swapped.

These are hinge questions for mistakes that come up again and again at GCSE. The topics are the ones pupils meet on AQA and Edexcel courses. They are not past papers, and they are not a mark scheme. Named texts, case studies and optional units differ by school. Swap the example. Keep the mistake.

Each wrong answer is a reason to reteach. The shaded box is for you. Cover it if the page is on the board.

36 hinges.

FRACTIONS · 1

What is 1/2 + 1/3?

A. 2/5
B. 5/6
C. 2/6
D. 1/6

Correct: B.

A. The numerators were added, and so were the denominators.

C. The numerators were added and a denominator was kept.

D. The fractions were multiplied instead of added.

Reteach: Use a common denominator: 3/6 + 2/6 = 5/6. Adding the bottoms is not how addition works.

FRACTIONS · 2

What is 1/2 × 1/3?

A. 1/6
B. 5/6
C. 2/5
D. 2/3

Correct: A.

B. The fractions were added. Multiplying by a fraction less than 1 should get smaller, not bigger.

C. The tops and the bottoms were added.

D. The numerators were added and the 3 was kept.

Reteach: Multiply the numerators and multiply the denominators. 1/2 of 1/3 is 1/6.

FRACTIONS · 3

What is 1/2 ÷ 1/4?

A. 2
B. 1/8
C. 2/4
D. 1/4

Correct: A.

B. The fractions were multiplied.

C. The tops were divided and the bottoms were divided.

D. The divisor was copied as the answer.

Reteach: Dividing by 1/4 is the same as multiplying by 4. How many quarters are in a half? Two.

NEGATIVE NUMBERS · 4

What is −3 − (−5)?

A. 2
B. −8
C. −2
D. 8

Correct: A.

B. The two negatives were added as if both were debts.

C. 5 was taken from 3 and the minus sign was kept.

D. The signs were dropped and the numbers added.

Reteach: Subtracting a negative is the same as adding. −3 + 5 = 2.

NEGATIVE NUMBERS · 5

What is −4 × −3?

A. 12
B. −12
C. −7
D. 1

Correct: A.

B. The sign of the first number was kept. Negative times negative is positive.

C. The numbers were added and a minus was attached.

D. The numbers were subtracted.

Reteach: Same signs give a positive product. 4 × 3 = 12, and two minuses make a plus.

DECIMALS · 6

What is 0.3 × 0.2?

A. 0.06
B. 0.6
C. 0.5
D. 6

Correct: A.

B. Only one decimal place was counted. The question has two.

C. The decimals were added.

D. The decimal points were ignored.

Reteach: 3 × 2 = 6, and there are two digits after the points in the question, so the answer is 0.06.

PERCENTAGES · 7

What is 20% of £50?

A. £10
B. £20
C. £30
D. £70

Correct: A.

B. 20% was read as £20.

C. 20 was subtracted from 50.

D. 20 was added to 50.

Reteach: 20% is one fifth. £50 ÷ 5 = £10. Percent means out of 100, not “the number of pounds”.

PERCENTAGES · 8

A price rises by 10%, then falls by 10%. The final price is

A. lower than the starting price
B. the same as the starting price
C. higher than the starting price
D. impossible to compare without knowing the starting price

Correct: A.

B. The two 10%s were treated as if they cancel.

C. The rise was treated as the last and larger change.

D. The result does not depend on the start. The fall is always the larger pound amount.

Reteach: The 10% fall is worked out on the higher price, so it removes more than the rise added. £100 becomes £110, then £99.

RATIO · 9

Red and blue counters are in the ratio 2:3. What fraction of the counters are red?

A. 2/5
B. 2/3
C. 3/5
D. 1/2

Correct: A.

B. The ratio was read as a fraction. 2:3 compares parts. It is not "2 out of 3".

C. The blue share was given.

D. The two numbers were treated as equal.

Reteach: Add the parts first. 2 + 3 = 5, so red is 2 of those 5 parts, which is 2/5.

RATIO · 10

£60 is shared in the ratio 2:3. The two shares are

A. £24 and £36
B. £2 and £3
C. £30 and £30
D. £20 and £40

Correct: A.

B. The parts of the ratio were treated as the pounds.

C. The money was halved and the ratio was ignored.

D. The 2 was ignored and the money was split using 3 only.

Reteach: 2 + 3 = 5 parts. One part is £60 ÷ 5 = £12. Then 2 × £12 and 3 × £12.

INDICES · 11

Simplify $y^3 \times y^4$.

A. y^7
B. y^{12}
C. y
D. y^{34}

Correct: A.

B. The powers were multiplied.

C. The powers were cancelled.

D. The powers were written next to each other.

Reteach: Same base, and you are multiplying: add the powers. 3 + 4 = 7. Multiply powers only when the power is raised again, as in $(y^3)^4$.

INDICES · 12

What is 5°?

A. 1
B. 0
C. 5
D. 50

Correct: A.

B. A power of zero was read as “nothing”.

C. The power was ignored.

D. The 0 was written on the end of the 5.

Reteach: Any non-zero number to the power 0 is 1. It is not 0. $5^1 = 5$, and each step down divides by 5, so $5^o = 1$.

STANDARD FORM · 13

Which of these is 4500 in standard form?

A. 4.5×10^3
B. 45×10^2
C. 0.45×10^4
D. 4.5×10^4

Correct: A.

B. This equals 4500, but standard form allows only one non-zero digit before the decimal point.

C. This equals 4500, but the first number has to be at least 1 and less than 10.

D. The power is one too big. That is 45000.

Reteach: Standard form is $a \times 10^n$, where a is from 1 up to but not including 10. $4500 = 4.5 \times 10^3$.

ALGEBRA · 14

Expand 3(x + 4).

A. 3x + 12
B. 3x + 4
C. 3x + 7
D. x + 12

Correct: A.

B. Only the first term inside the bracket was multiplied.

C. 3 was added to 4 instead of multiplying.

D. Only the 4 was multiplied by 3.

Reteach: The number outside multiplies every term inside. 3 × x and 3 × 4.

ALGEBRA · 15

Expand $(x + 5)^2$.

A. $x^2 + 10x + 25$
B. $x^2 + 25$
C. $x^2 + 5x + 25$
D. $x^2 + 10x + 10$

Correct: A.

B. Each term was squared and the middle term was dropped.

C. The middle term was not doubled.

D. 5 was doubled instead of squared.

Reteach: $(x + 5)^2$ means $(x + 5)(x + 5)$. The middle term is 2 × x × 5. Square both ends and double the middle.

ALGEBRA · 16

Factorise $x^2 + 7x + 12$.

A. (x + 3)(x + 4)
B. (x + 7)(x + 12)
C. (x + 2)(x + 6)
D. (x + 1)(x + 12)

Correct: A.

B. The two numbers already written in the quadratic were put in the brackets.

C. The numbers multiply to 12, but they add to 8, not 7.

D. The numbers multiply to 12, but they add to 13, not 7.

Reteach: Find two numbers that multiply to 12 and add to 7. That is 3 and 4. Check by expanding.

EQUATIONS · 17

Solve 2x + 6 = 14.

A. x = 4
B. x = 10
C. x = 8
D. x = 7

Correct: A.

B. 6 was added to 14 instead of subtracted, then the result was halved.

C. 6 was subtracted, then the divide by 2 was forgotten. 14 − 6 is not the value of x.

D. 14 was divided by 2 and the + 6 was ignored.

Reteach: Undo the + 6 first: 2x = 8. Then divide by 2. Check: 2 × 4 + 6 = 14.

INEQUALITIES · 18

Solve −2x > 8.

A. x < −4
B. x > −4
C. x < 4
D. x > 4

Correct: A.

B. Both sides were divided by −2, and the inequality sign was not reversed.

C. The sign of the 4 was dropped when the inequality was reversed.

D. 8 was divided by 2 and both signs were left as they were.

Reteach: Dividing by a negative reverses the inequality. −2x > 8 becomes x < −4. Check with a number: x = −5 works, x = −3 does not.

SEQUENCES · 19

The sequence is 3, 7, 11, 15. What is the nth term?

A. 4n − 1
B. 4n + 3
C. n + 4
D. 3n + 4

Correct: A.

B. The first term was used as the number you add. For n = 1 that gives 7, not 3.

C. The difference was added to n. That does not generate the sequence.

D. The first term was used as the multiplier.

Reteach: The difference is 4, so start from 4n. For n = 1 you need 3, and 4 × 1 − 1 = 3. Always test n = 1.

GRAPHS · 20

A line passes through (0, 1) and (2, 5). What is the gradient?

A. 2
B. 5
C. 4
D. 1/2

Correct: A.

B. The y-coordinate of the second point was used.

C. The change in y was found and not divided by the change in x.

D. Run was divided by rise, instead of rise by run.

Reteach: Gradient is change in y divided by change in x. (5 − 1) / (2 − 0) = 2.

GRAPHS · 21

Where does y = 3x + 2 cross the y-axis?

A. (0, 2)
B. (0, 0)
C. (2, 0)
D. (0, 3)

Correct: A.

B. The line was assumed to pass through the origin.

C. + 2 was read as the x-intercept.

D. The gradient was used instead of the intercept.

Reteach: In y = mx + c, c is the y-intercept. When x = 0, y = 2, so the crossing point is (0, 2).

AREA · 22

A triangle has base 8 cm and perpendicular height 5 cm. What is its area?

A. 20 cm²
B. 40 cm²
C. 13 cm²
D. 4 cm²

Correct: A.

B. The half was forgotten. Base × height is the rectangle, not the triangle.

C. The base and the height were added.

D. The base was halved and the height was ignored.

Reteach: Area of a triangle is ½ × base × perpendicular height. The height has to be at right angles to the base you used.

AREA AND PERIMETER · 23

A rectangle is 6 cm by 4 cm. Which pair is right?

A. Area 24 cm², perimeter 20 cm
B. Area 20 cm², perimeter 24 cm
C. Area 24 cm², perimeter 10 cm
D. Area 10 cm², perimeter 24 cm

Correct: A.

B. Area and perimeter were swapped.

C. The two sides were added once and the trip around was not finished.

D. The sides were added for the area and multiplied for the perimeter.

Reteach: Area is length × width, in cm². Perimeter is the distance round: 6 + 4 + 6 + 4 = 20 cm.

CIRCLES · 24

A circle has radius 3 cm. What is its area?

A. 9π cm²
B. 6π cm²
C. 3π cm²
D. 36π cm²

Correct: A.

B. That is the circumference, 2πr, not the area.

C. π was multiplied by the radius once.

D. The diameter was used in πr².

Reteach: Area is πr². Circumference is 2πr. 6π is the distance around this circle, not the space inside it.

PYTHAGORAS · 25

A right-angled triangle has hypotenuse 10 cm and one shorter side 6 cm. The other shorter side is

A. 8 cm
B. √136 cm
C. 4 cm
D. 16 cm

Correct: A.

B. The squares were added, which is what you do when the hypotenuse is the missing side.

C. The lengths were subtracted.

D. The lengths were added.

Reteach: The hypotenuse is the longest side. To find a shorter side, subtract the squares: $\sqrt{(10^2 - 6^2)} = 8$. Add the squares only when you are finding the hypotenuse.

TRIGONOMETRY · 26

In a right-angled triangle the opposite is 4 cm and the hypotenuse is 8 cm. 4/8 is

A. sin of the angle
B. cos of the angle
C. tan of the angle
D. impossible to know without the adjacent

Correct: A.

B. Opposite over hypotenuse was called cos.

C. Opposite over hypotenuse was called tan. Tan uses the adjacent, not the hypotenuse.

D. The adjacent is not needed for this ratio.

Reteach: SOHCAHTOA: sin = opposite / hypotenuse. Cos uses the adjacent. Tan uses opposite and adjacent, and no hypotenuse.

ANGLES · 27

Two angles sit on a straight line. One is 70°. The other is

A. 110°
B. 290°
C. 90°
D. 20°

Correct: A.

B. 360° was used. That is a full turn, not a straight line.

C. The angle was assumed to be a right angle.

D. 70° was subtracted from 90° instead of from 180°.

Reteach: Angles on a straight line add to 180°, not 360°. 180 − 70 = 110.

ANGLES · 28

What is the sum of the interior angles of a hexagon?

A. 720°
B. 1080°
C. 360°
D. 120°

Correct: A.

B. The number of sides was multiplied by 180. The rule is (n − 2) × 180.

C. The sum of the exterior angles was used as if it were the interior sum.

D. One interior angle of a regular hexagon was given, not the sum.

Reteach: Interior sum = (n − 2) × 180°. For 6 sides that is 4 × 180 = 720°. 120° is one angle only if the hexagon is regular.

PROBABILITY · 29

A pupil says the probability of an event is 1.2. What is right?

A. 1.2 cannot be a probability
B. 1.2 means the event is very likely
C. 1.2 means the event is certain
D. 1.2 means it happens 12 times out of 10, which is allowed

Correct: A.

B. Numbers above 1 were treated as “even more certain”.

C. Certain is 1. Anything above 1 is off the scale.

D. A probability cannot be greater than 1.

Reteach: Probabilities run from 0 to 1 inclusive. 0 is impossible, 1 is certain. 1.2 is not on the scale.

PROBABILITY · 30

A fair spinner can land only on red or on blue, with equal chance. What is the probability it lands on red or blue?

A. 1
B. 1/4
C. 1/2
D. 0

Correct: A.

B. The two probabilities were multiplied. That is a method for “and”, not for this “or”.

C. The probability of one colour was given.

D. “Or” was read as “both at once”, which this spinner cannot do.

Reteach: Red and blue cover every outcome, so the probability is 1. Multiply for independent events joined by “and”. Add outcomes that do not overlap when the word is “or”.

AVERAGES · 31

The numbers are 2, 3 and 7. What is the mean?

A. 4
B. 3
C. 7
D. 2

Correct: A.

B. The median, the middle value, was given.

C. The largest value was given.

D. The first value was given.

Reteach: The mean is the total divided by how many. (2 + 3 + 7) ÷ 3 = 4. The median of this list is 3. They are not the same.

CORRELATION · 32

Ice-cream sales and cases of sunburn both rise in summer. What is a sound conclusion?

A. They rise together. That does not mean ice cream causes sunburn.
B. Ice cream causes sunburn
C. Sunburn causes people to buy ice cream
D. There is no link of any kind

Correct: A.

B. A link was treated as one thing causing the other.

C. The link was turned into a cause, the other way round.

D. The pattern was denied because it is not a cause.

Reteach: The two can rise together because hot weather affects both. Correlation is not the same as one variable causing the other.

BOUNDS · 33

A length is 7 cm, measured to the nearest centimetre. The lower bound is

A. 6.5 cm
B. 6 cm
C. 6.9 cm
D. 7 cm

Correct: A.

B. A whole centimetre was taken off.

C. One tenth was taken off, which would suit a measurement to the nearest tenth.

D. The rounded value was treated as the bound.

Reteach: To the nearest centimetre, 7 covers from 6.5 cm up to but not including 7.5 cm. The lower bound is 6.5 cm.

COMPOUND MEASURES · 34

A car travels 120 miles in 2 hours. What is its average speed?

A. 60 mph
B. 240 mph
C. 122 mph
D. 118 mph

Correct: A.

B. Distance and time were multiplied.

C. The numbers were added.

D. The numbers were subtracted.

Reteach: Speed = distance ÷ time. 120 ÷ 2 = 60. Units: miles per hour.

COMPOUND MEASURES · 35

A block has mass 40 g and volume 8 cm³. What is its density?

A. 5 g/cm³
B. 0.2 g/cm³
C. 320 g/cm³
D. 48 g/cm³

Correct: A.

B. Volume was divided by mass. Density is mass divided by volume.

C. Mass and volume were multiplied.

D. Mass and volume were added.

Reteach: Density = mass ÷ volume. 40 ÷ 8 = 5 g/cm³. A bigger number means more mass in each cubic centimetre.

ENLARGEMENT · 36

A shape is enlarged by scale factor 3. What happens to its area?

A. The area is multiplied by 9
B. The area is multiplied by 3
C. The area is multiplied by 6
D. The area stays the same

Correct: A.

B. The length scale factor was used for the area.

C. The scale factor was doubled.

D. Enlargement was treated as a move, not a change of size.

Reteach: Area scale factor is the length scale factor squared. $3^2 = 9$. Lengths × 3, area × 9.

Run one hinge in two minutes

The questions in this bank are multiple choice, so a class can show A, B, C or D on mini whiteboards. That is the fast version. You see the letter, and you reteach the line that goes with it.

If you want the sentence behind the letter, use Get It?.

1. Type the topic, or paste the question. Edit what comes back. It is your question.
2. Pupils join with the class code on a phone, a laptop, or a school device. A first name is enough. No account, and nothing to install.
3. They answer in their own words, typed or spoken. It is not multiple choice. The sentence shows the mistake more clearly than a letter does.
4. In about two minutes you get green, amber or red. The colour is guidance. It can be wrong, and you can change it.
5. Reteach the line from this bank before you move on.

Where phones are banned, do not demand a phone. Use school devices, or stay with the boards.

A first name and the answers are sent to an external AI. If school policy forbids that, do not use the tool with pupil answers. The boards still work.

Free during the beta. No card.

dotheygetit.app